Equipment list + datasheets

SO3 Converter Ver 1 · SO3-V1-ME-EQL-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

REV	DATE	DESCRIPTION	ORIGINATOR	CHECKED / APPROVED
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Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose and scope

This document lists every item of equipment of the SO3 Converter Ver 1 converter section (battery limits: dried air in, molten sulfur in, acid and steam out, tail gas to stack) with its service, type, size, operating and design conditions, material and code, and gives datasheets for the six major items. It is generated from the simulator model at its design point, the Design tab (R-201 diameter 13.0 m assumed), PFD-201 and the EPC P&IDs SO3-PID-001/002 rev A. Items outside the model (sulfur pit, filters, acid coolers, stack) are listed for the interfaces and marked assumed.

2 · Design basis

- Design temperature = highest credible operating temperature including alarm and trip limits + 25 K, rounded up to 5 °C (typical practice).
- Design pressure, liquids and steam = max(operating + 10 %, operating + 170 kPa), rounded up to 10 kPa (typical practice).
- Design pressure, gas train K-101 to stack = 1.1 × K-101 shut-off pressure (75 kPa g, model blower curve), rounded up: 100 kPa g; vacuum −5 kPa g (assumed, hot shutdown cooling).
- Pump casings: design pressure ≥ shut-off (120 % of rated head assumed; molten sulfur on the model supply pressure 600 kPa).
- R-201 pass 1, SH-201, E-201: 700 °C (TSHH-203 660 °C + margin; the SRS assumes the same 700 °C).
- Operating pressures after bed 1 = model bed ΔP (PDI-204…244) plus the other losses (PI-200 − Σ bed ΔP) split in assumed shares.
- Codes quoted as typical: ASME VIII Div 1, ASME I, TEMA R, API 610, API 617/672, NFPA 85/86. Materials typical for the service.

Operating pressure profile, kPa g:

K-101	F-101	Bed 1 in	Bed 1 out	Bed 2 in	Bed 3 in	T-301 in	Bed 4 in	Bed 5 in	T-302 in	Stack
60.0	41.0	35.0	30.5	28.2	23.0	18.0	11.5	8.0	3.7	0.0

3 · Equipment list

Tag	Service	Type	Size / duty	Op. P kPa g	Des. P kPa g	Op. T °C	Max op. T °C (basis)	Des. T °C	Material	Code	Dwg
K-101	Main air blower	Centrifugal blower, single stage, motor driven (typical)	238.5 kNm³/h (308 t/h); Δp 61 kPa; 4200 kW shaft	60.0	100	90	110 (hot afternoon +20 °C (model fault case))	135	Casing carbon steel, impeller high-strength alloy steel (typical)	API 672 / API 617 (typical)	SO3-PID-001
F-101	Sulfur furnace	Horizontal refractory-lined cylinder, sulfur guns, baffle walls	ID 5 m × 12 m (PFD-201); 104.6 MW released	41.0	100	1121	300 (shell metal behind refractory (assumed); gas up to the adiabatic flame 1161 °C on the refractory)	325 (shell metal; refractory hot face 1400 °C (typical))	Carbon steel shell SA-516-70, 3-layer refractory (typical)	ASME VIII-1 shell; refractory to supplier (typical)	SO3-PID-001
H-101	Preheat burner (start-up only)	Fuel-gas burner package with BMS, igniter and scanner	30 MW firing (model maximum)	41.0	100	—	—	package	Package (vendor standard)	NFPA 85/86, EN 746-2 (typical)	SO3-PID-001
B-101	Waste heat boiler	Fire-tube boiler, horizontal, natural circulation to steam drum	Shell 3.5 m × 9 m (PFD-201); 61.6 MW; 94 t/h steam	4000	4400	250	250	280	SA-516-70 shell, SA-192/SA-210 tubes (typical)	ASME I or ASME VIII-1 (unfired steam boiler) per jurisdiction	SO3-PID-001
R-201	SO_2 converter, 5 adiabatic beds (3+2)	Vertical cylindrical, flat beds on grids, division plates, central/side ducts	ID 13.0 m × 16.6 m T/T (assumed); catalyst 340 m³	35.0	100	631	660 (TSHH-203 trip 660 °C (alarm TI-203 640 °C))	700 (pass 1; passes 2–5 per zone)	304H stainless steel shell and grids (typical; older plants carbon steel brick-lined)	ASME VIII-1	SO3-PID-001/002
SH-201	Steam superheater (bed 1 gas)	Shell-and-tube, gas in shell, steam in tubes	2.24 MW; steam 250 → 292 °C	4000	4400	631	660 (bed 1 exit gas: TSHH-203 660 °C)	700	304H shell; tubes SA-213 TP304H (typical)	ASME VIII-1 / TEMA R	SO3-PID-001
E-201	Hot interpass exchanger (gas–gas)	Shell-and-tube, single pass, counter-current, TV-211 bypass	17.2 MW; UA 89 kW/K; ≈ 3570 m² at U 25 W/m² K (assumed)	28.2	100	610	660 (hot inlet can see bed 1 exit gas (TV-231 open): 660 °C)	700	304H / 321 stainless steel (typical)	ASME VIII-1 / TEMA R	SO3-PID-001
E-202	Cold interpass exchanger (gas–gas)	Shell-and-tube, counter-current, TV-221 bypass	9.2 MW; UA 27.8 kW/K; ≈ 1120 m² at U 25 W/m² K (assumed)	23.0	100	528	543 (bed 2 exit with bed 2 inlet at TIC-211 high alarm 460 °C (+15 K))	570	304 SS; stainless cold end against acid condensation (typical)	ASME VIII-1 / TEMA R	SO3-PID-001
E-203	Economizer before IPA	Finned-tube economizer, BFW in tubes, TV-251 gas bypass	27.9 MW; BFW 150 t/h	5000	5500	465	480 (bed 3 exit with bed 3 inlet at TIC-221 high alarm 455 °C)	510	Carbon steel tubes, stainless cold end (typical)	ASME VIII-1 / ASME I economizer	SO3-PID-002
E-204	Interbed economizer (bed 4 → 5)	Finned-tube economizer, TV-241 gas bypass	1.03 MW	5000	5500	441	461 (bed 4 exit with bed 4 inlet at TIC-231 high alarm 445 °C)	490	Carbon steel / stainless (typical)	ASME VIII-1	SO3-PID-002
E-205	Final economizer before FAT	Finned-tube economizer, no bypass	20.8 MW	5000	5500	427	445 (bed 5 at TIC-241 high alarm 445 °C)	470	Carbon steel, stainless cold end (typical)	ASME VIII-1	SO3-PID-002
T-301	Intermediate absorber (IPA)	Packed tower, acid distributor, candle mist eliminators	ID 7.1 m at 2.3 m/s (assumed); acid 1600 m³/h	18.0	100	190	230 (gas in at TIC-251 high alarm 230 °C)	255	Carbon steel brick-lined or high-Si austenitic stainless (typical)	ASME VIII-1 (low pressure)	SO3-PID-002
T-302	Final absorber (FAT)	Packed tower, candle mist eliminators	ID 6.9 m at 2.3 m/s (assumed); acid 1000 m³/h	3.7	100	160	185 (gas in: no alarm on TI-252; +25 K allowance (assumed))	215	Carbon steel brick-lined or high-Si austenitic stainless (typical)	ASME VIII-1 (low pressure)	SO3-PID-002
TK-301	IPA pump tank	Horizontal tank, acid return and pump wells	≈ 140 m³ (5 min circulation, assumed)	0.0	atm + liquid head	80	95 (TI-304 high alarm 95 °C)	120	Carbon steel brick-lined / alloy (typical)	API 650 / ASME VIII-1 (typical)	SO3-PID-002
TK-302	FAT pump tank	Horizontal tank	≈ 90 m³ (5 min circulation, assumed)	0.0	atm + liquid head	80	95 (no alarm; +15 K (assumed))	120	Carbon steel brick-lined / alloy (typical)	API 650 / ASME VIII-1 (typical)	SO3-PID-002
P-101A/B	Molten sulfur pumps (A run, B spare by hand)	Vertical submerged, steam-jacketed (API 610 VS4, typical)	22.9 m³/h at Δp 600 kPa (model supply pressure)	600	1100	140	155 (steam tracing at 0.45 MPa g, 155 °C (assumed))	180	Carbon steel casing, steam jacket (typical)	API 610 (typical)	SO3-PID-001

Numbers from the simulator model, design point, noise off; "assumed" where stated.

Tag	Service	Type	Size / duty	Op. P kPa g	Des. P kPa g	Op. T °C	Max op. T °C (basis)	Des. T °C	Material	Code	Dwg
FL-101A/B	Molten sulfur filters (tag assumed, not on the P&ID)	Steam-jacketed pressure leaf filter, duty/standby (typical)	22.9 m³/h	600	1100	140	155 (as P-101)	180	Carbon steel, SS leaves (typical)	ASME VIII-1	— (HAZOP node 1 text)
TK-101	Molten sulfur pit (outside battery limit)	Concrete pit, steam coils, vents (typical)	≈ 550 m³ (24 h working, assumed)	0.0	atm	140	155 (steam coils)	180	Acid-resistant lined concrete (typical)	by others	SO3-PID-001 (terminal)
P-301A/B	IPA acid pumps (B auto-start HS-301B)	Vertical cantilever acid pump in TK-301 (typical)	1600 m³/h × 22 m (assumed); 225 kW absorbed	380	700	80	95 (TI-304 high alarm 95 °C)	120	High-Si stainless / alloy wetted parts (typical)	API 610 VS (typical)	SO3-PID-002
P-302A/B	FAT acid pumps (B manual)	Vertical cantilever acid pump in TK-302 (typical)	1000 m³/h × 22 m (assumed); 141 kW absorbed	380	700	80	95 (+15 K (assumed))	120	High-Si stainless / alloy (typical)	API 610 VS (typical)	SO3-PID-002
E-301/E-302	IPA / FAT acid coolers (acid system package, tags assumed)	Plate or anodically protected shell-and-tube (typical)	55.2 / 7.9 MW (absorption heat 132 kJ/mol assumed + gas)	400	700	105	120 (acid from tower ≈ TI-304 + 25 K (assumed))	145	316L / alloy C-276 plates; anodic protection (typical)	ASME VIII-1	— (acid system P&ID, note 2)
ST-301	Stack	Self-supporting stack with liner	ID 2.5 m at 15 m/s (assumed); 195.7 kNm³/h; height by dispersion study	0.5	atm + draft	80	160 (FAT bypassed: gas at TI-252)	190	FRP or stainless-lined steel (typical)	CICIND / EN 13084 (typical)	SO3-PID-002

4 · Datasheets

4.1 · Datasheet R-201

Item	Value	Unit	Source
Service	SO_2 converter, 5 adiabatic beds, double absorption after bed 3		PFD-201
Inside diameter	13.0	m	Design tab DS_DREF (assumed)
Height tangent to tangent	16.6	m	beds + 0.15 m ball layers + 2 m gas spaces + plenums (assumed)
Catalyst beds 1–5	61.1 / 68.0 / 82.0 / 68.0 / 61.1	m^3	Design tab dsCalc (K.Wb / 0.55 t/m^3)
Catalyst mass 1–5	33.6 / 37.4 / 45.1 / 37.4 / 33.6	t	model K.Wb
Bed depths 1–5	0.46 / 0.51 / 0.62 / 0.51 / 0.46	m	Design tab (13.0 m diameter)
Gas flow into bed 1	238.5 kNm^3/h; 348 t/h		model FI-100, gas1
Superficial velocity	0.50	Nm/s	Design tab
Inlet / outlet °C by bed	420→631 · 445→528 · 440→465 · 425→441 · 428→427	°C	model design state
Pass ΔP 1–5	4.52 / 2.86 / 3.16 / 2.54 / 2.50	kPa	model PDI-204…244
Operating pressure (bed 1 inlet)	35.0	kPa g	PI-200
Design pressure	100 / −5	kPa g	≥ 1.1 × K-101 shut-off 75 kPa (model K.H0), rounded
Design temperature pass 1 / 2 / 3–5	700 / 570 / 490	°C	TSHH-203 660 °C + margin; SRS assumes 700 °C
Heat of reaction released	34.87	MW	model Qconv
Material	304H stainless steel shell, division plates and grids		typical
Corrosion allowance	0 (stainless) / 3 mm if carbon steel	mm	typical
Code	ASME VIII Div 1; insulation external, personnel protection		typical

4.2 · Datasheet B-101

Item	Value	Unit	Source
Service	Waste heat boiler after F-101, hot-gas bypass TV-201		PFD-201
Type	Fire-tube, horizontal, natural circulation to steam drum		P&ID note 7
Shell size	3.5 m × 9 m		PFD-201 sketch size
Gas in / out	1121 / 274	°C	TI-106 / TI-107
Gas through tubes (1 − bypass)	79.3 % of furnace gas		TV-201 34.6 % × 0.6
NTU (model)	3.57	—	Design tab
Duty	61.6	MW	Design tab = FI-108 × (2801 − 440) kJ/kg
Steam	94.0 t/h saturated, 4 MPa g, 250 °C (model)		FI-108, K.Tsat
Design pressure steam / gas	4.4 MPa g / 100 kPa g		op + 10 % (typical)
Design temperature steam side	280	°C	Tsat + 30 K (typical)
Safety valves	PSV-101 on drum / B-101 steam side, set ≤ 4.4 MPa g		Safety-layer register (recommended)
Materials	SA-516-70 shell and tubesheets, SA-192/SA-210 tubes, refractory + ferrules at gas inlet		typical
Code	ASME I or ASME VIII-1 (unfired steam boiler)		typical

4.3 · Datasheet E-201

Item	Value	Unit	Source
Service	Hot interpass: bed 1 gas (after SH-201) heats return gas from E-202 to the bed 4 inlet		PFD-201
Hot side in / out (through tubes)	610.3 / 405.8	°C	model (TV-231 mix; xE201h)
Cold side in / out	203.6 / 425.0	°C	TI-254 / TI-231
Heat capacity rates Ch / Cc	84.3 / 77.8	kW/K	model cpMix × flow (1 − TV-211 bypass)
TV-211 bypass at design	19.2	% of hot gas	opening 32.0 % × 0.6
Duty	17.23	MW	model q201
UA installed / minimum (no bypass)	89.0 / 81.4	kW/K	model K.UA201 corrected / Design tab
NTU, Cr, ε	1.14, 0.92, 0.544		ε–NTU counter-current
Overall U (assumed)	25	W/m² K	gas–gas, typical 15–35
Area	3570	m²	UA / U
Design P / T	100 kPa g / 700 °C		as R-201 pass 1
Material	304H / 321 stainless steel tubes and shell		typical
Code	ASME VIII-1, TEMA R; expansion bellows on shell		typical

4.4 · Datasheet T-301

Item	Value	Unit	Source
Service	Intermediate (interpass) absorber: SO_3 into 98.5 % acid		PFD-201
Gas in	224.8 kNm³/h at 190 °C; SO_3 12.20 %		Design tab nIpa, TIC-251
Actual gas flow	89.9	m³/s	at 18.0 kPa g
Superficial velocity (assumed)	2.3	m/s	typical 2.0–2.7 for acid towers
Inside diameter	7.1	m	√(4Q / π u), rounded up
Acid circulation	1600 m³/h, 98.5 % H_2SO_4, 80 °C		model FI-301, AI-303, TI-304
Irrigation rate	40.4	m³/m² h	flow / area
SO_3 absorbed	97.9 t/h (η 99.95 %)		Design tab abs1, K.eta1
Gas out	83 °C		TI-305
Packing (assumed)	4.5 m ceramic saddles on ceramic grid; candle mist eliminators		typical
Design P / T	100 kPa g / 255 °C		TIC-251 high alarm 230 °C + 25 K
Material	Carbon steel shell, acid-resistant brick lining; or high-Si austenitic stainless		typical
Code	ASME VIII-1		typical

4.5 · Datasheet K-101

Item	Value	Unit	Source
Service	Main air blower: dried air to F-101 (FV-101) and dilution (FV-102)		PFD-201
Flow	238.5 kNm³/h = 85.7 kg/s		FI-100 (1.293 kg/Nm³)
Suction (assumed)	-1 kPa g, 45 °C (after drying tower)		assumed
Discharge	60.0 kPa g, 90 °C		PI-101, model air temperature
Shut-off pressure	75	kPa g	model blower curve H0
Shaft power (model)	4200	kW	econ() / Efficiency tab
Isentropic power (hand check)	3982	kW	k = 1.4, pressure ratio 1.608
Implied isentropic efficiency	0.95	—	typical 0.80–0.85 → see EQ
Isentropic discharge T	91.2	°C	matches the model 90 °C only at η ≈ 1
Driver	Induction motor 6.6 kV (Electrical tab); steam turbine is the usual alternative		typical
Protection	Anti-surge / minimum flow, non-return damper, vibration and bearing T		HAZOP R12
Code	API 672 / API 617 (typical)		typical

4.6 · Datasheet P-301A

Item	Value	Unit	Source
Service	IPA acid circulation, P-301B identical spare with auto-start (HS-301B)		P&ID 002, model
Liquid	98.5 % H_2SO_4, 80 °C, ρ 1760 kg/m³ (typical)		model AI-303, TI-304
Rated flow	1600	m³/h	FI-301
Head (assumed)	22	m	tower distributor height + cooler + losses
Differential pressure	380	kPa	ρ g H
Hydraulic power	169	kW	Q Δp
Efficiency (assumed)	0.75	—	large vertical pump, typical
Absorbed power	225	kW	to Electrical load list
Shut-off (assumed 120 % of rated head)	456	kPa	typical curve
Low-flow protection	FSLL-301 < 800 m³/h trips sulfur; auto-start of B below 60 %		model
Type / material	Vertical cantilever, high-Si stainless or alloy wetted parts		typical
Code	API 610 VS (typical)		typical

5 · Shell thickness, ASME VIII Div 1

Cylinder (UG-27): t = P R / (S E − 0.6 P) + CA; handling minimum t ≥ D/1000 + 2.5 mm (typical). Allowable stresses are approximate teaching values in the shape of ASME II-D; check the current edition for design. R-201 at 100 kPa g, 13.0 m ID, 304H, E 0.85, CA 0:

Zone	Design T °C	S 304H MPa	t pressure mm	t minimum mm	Governs	Nominal mm
Pass 1	700	26.9	28.5	15.5	pressure	29
Pass 2	570	69.6	11.0	15.5	minimum (handling)	16
Passes 3–5	490	73.7	10.4	15.5	minimum (handling)	16

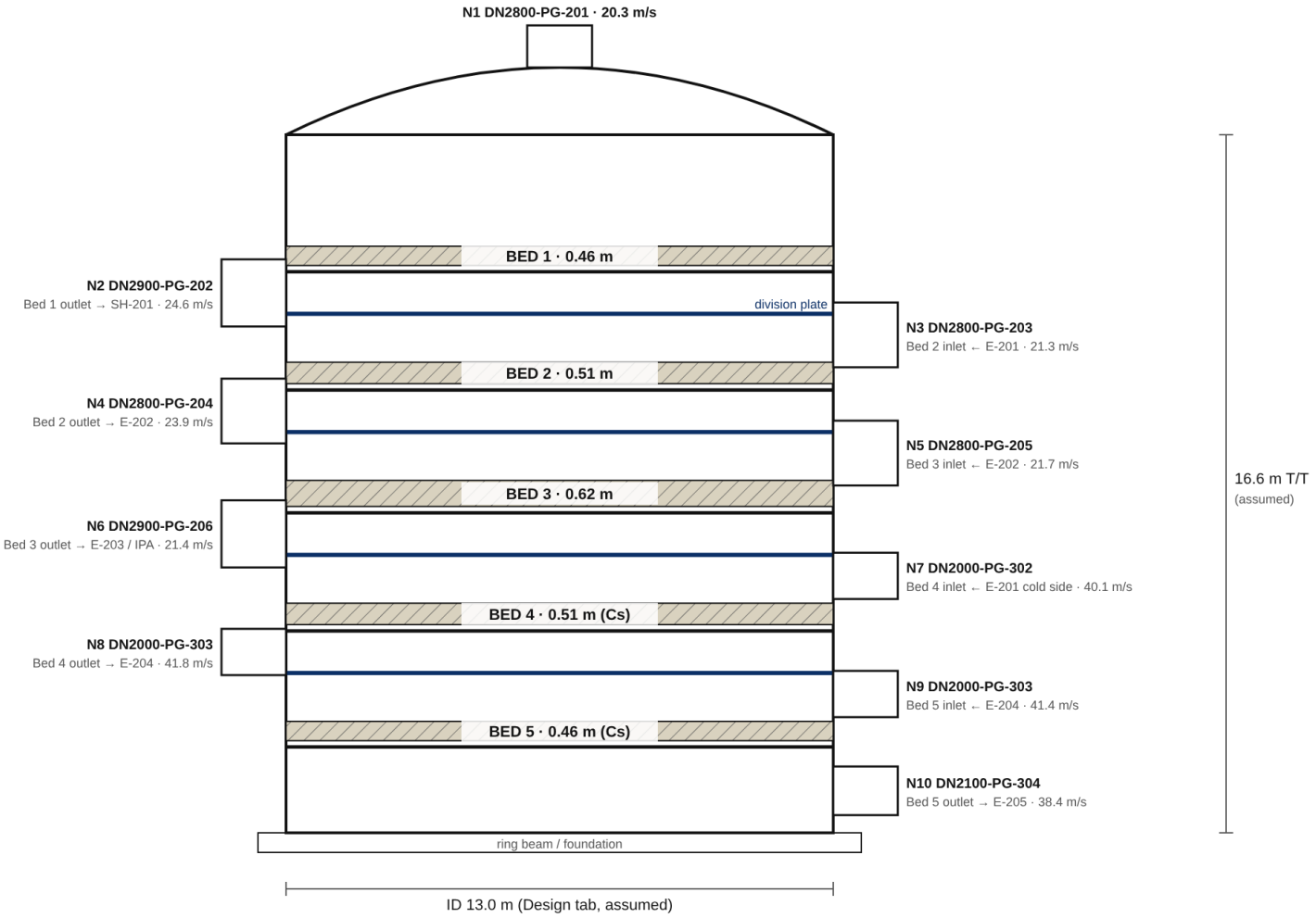
At 100 kPa g and 700 °C, 304H (S ≈ 27 MPa, E 0.85) needs 28.5 mm against a handling minimum of 15.5 mm; passes 2–5 need 11.0 mm (minimum governs). "Near-atmospheric, so the minimum governs" holds only below about 631 °C for this shell.

6 · Interpass exchanger rating (ε–NTU)

Quantity	E-201	E-202
Hot in °C	610.3	528.0
Hot out °C	405.8	416.4
Cold in °C	203.6	83.0
Cold out °C	425.0	203.6
Ch through exchanger kW/K	84.3	82.1
Cc kW/K	77.8	75.9
Hot bypass % of gas	19.2	21.1
UA kW/K	89.0	27.8
UA minimum, no bypass kW/K	81.4	27.0
NTU	1.144	0.367
Cr	0.924	0.925
ε	0.544	0.271
Q kW (model)	17235	9157
U assumed W/m² K	25	25
Area m²	3570	1120

Counter-current ε–NTU with the model's UA correction for flow; the hot bypass (TV-211/TV-221) takes gas round the exchanger, so the installed UA must exceed the no-bypass minimum.

7 · R-201 general arrangement and nozzles



Notes: elevation, not to scale for nozzles. Nozzle sizes = P&ID line sizes (SO3-PID-001/002); velocity = model gas flow at the design state. Manhole DN600 and catalyst drop-out above/below each bed, TC nozzles TI-2x1/2x2/2x3 and PDI taps per bed (typical, not drawn). HV-112 start-up line ties into PG-302 outside the vessel. Ball layers 0.15 m, gas spaces 2 m (assumed).

R-201 GENERAL ARRANGEMENT · ELEVATION · NTS
SO3-V1-ME-EQL-001 · Rev 0 · Issued for teaching

Nozzle	Service	Line	T °C	P kPa g	m³/s	m/s	DN at 22.5 m/s
N1	Bed 1 inlet (top)	DN2800-PG-201	420	35.0	125.0	20.3	DN2700
N2	Bed 1 outlet → SH-201	DN2900-PG-202	631	30.5	162.2	24.6	DN3100
N3	Bed 2 inlet ← E-201	DN2800-PG-203	445	28.2	131.1	21.3	DN2800
N4	Bed 2 outlet → E-202	DN2800-PG-204	528	25.3	147.3	23.9	DN2900
N5	Bed 3 inlet ← E-202	DN2800-PG-205	440	23.0	133.5	21.7	DN2800
N6	Bed 3 outlet → E-203 / IPA	DN2900-PG-206	465	19.8	141.2	21.4	DN2900
N7	Bed 4 inlet ← E-201 cold side (HV-112 ties in on the line)	DN2000-PG-302	425	11.5	125.9	40.1	DN2700
N8	Bed 4 outlet → E-204	DN2000-PG-303	441	9.0	131.3	41.8	DN2800
N9	Bed 5 inlet ← E-204	DN2000-PG-303 (return, assumed same size)	428	8.0	130.1	41.4	DN2800
N10	Bed 5 outlet (bottom) → E-205	DN2100-PG-304	427	5.5	133.0	38.4	DN2800

8 · Checks against the model and the other documents

14 of 21 checks pass. The same checks are recomputed live in the Mechanical tab of the simulator.

Check	Expected	Found	Result	Source
Gas-train design pressure ≥ K-101 shut-off (blower curve)	≥ 75 kPa g	lowest of 7 items: 100 kPa g	✓	model K.H0
B-101 steam-side design pressure ≥ PSV-101 set	≥ 4400 kPa g	4400 kPa g	✓	Safety-layer register
Pump casing design pressure ≥ shut-off (P-101, P-301, P-302)	margin ≥ 0	380 / 244 / 244 kPa	✓	K.Psup, curves assumed
R-201 pass 1 design T ≥ TSHH-203 trip + 25 K	≥ 685 °C	700 °C	✓	K.tripT, ALM TI-203 640 °C
R-201 pass 1 design T ≥ peak bed 1 exit with SIF-02 acting (SRS)	≥ 744.1 °C	700 °C	✗	SRS SIF-02 PST run → EQ-em-01
Every item: design T ≥ max operating T incl. alarm limits	21 items	all 21 pass	✓	ALM, K.tripT
T-301 design T ≥ TIC-251 high alarm	≥ 230 °C	255 °C	✓	ALM TIC-251
TK-301 / P-301 design T ≥ TI-304 high alarm	≥ 95 °C	120 / 120 °C	✓	ALM TI-304
B-101 duty on the datasheet = model steam × Δh	61.6 MW	61.6 MW	✓	FI-108
B-101 duty in the HAZOP / safety text (≈ 45 MW)	61.6 MW	45 MW	✗	HAZOP node 3, register → EQ-em-02
R-201 heat of reaction (Design tab) vs model	34.87 MW	34.87 MW	✓	P.Qconv
E-201 duty (Design tab) vs model q201	17.24 MW	17.22 MW	✓	P.q201
E-202 duty (Design tab) vs model q202	9.16 MW	9.18 MW	✓	P.q202
E-201 installed UA ≥ UA needed with no bypass	≥ 81.4 kW/K	89.0 kW/K (+9 %)	✓	Design tab
Every PFD-201 item is in the equipment list	13 PFD tags	all 13 found; +9 P&ID / auxiliary items	✓	pfd/pf_spec.json
R-201 datasheet diameter = Design tab diameter	13.0 m	13.0 m	✓	DS_DREF
P&ID rev A design-point table: bed ΔP vs model PDI-204…244	4.52/2.86/3.16/2.54/2.50 kPa	4.5/3.0/3.6/2.9/3.0 kPa (max diff 0.50)	✗	SO3-PID-001 → EQ-em-03
P&ID rev A note: stack SO_2 vs model AIT-306	292 ppm	286 ppm	✗	SO3-PID-001 → EQ-em-03
R-201 nozzle gas velocity in the P&ID range 15–30 m/s (note 2: ducts 20–25)	15–30 m/s	20.3–41.8 m/s	✗	P&ID line sizes
K-101 model power vs isentropic hand check (η typical 0.80–0.85)	η 0.80–0.85	η 0.95 (3982 kW isentropic / 4200 kW)	✗	econ(), EF_D → EQ-em-04
R-201 pass 1 shell: thickness set by the minimum, not by pressure (brief)	minimum governs	pressure: 28.5 mm vs min 15.5 mm	✗	ASME VIII-1 UG-27 → EQ-em-05

9 · Engineering queries

No.	Document	Subject	Finding	Proposed
EQ-em-01	SO3-V1-ME-EQL-001	R-201 pass 1, SH-201, E-201 design temperature vs SIF-02 response	Pass-1 design temperature is 700 °C (TSHH-203 660 °C + margin; the SRS also assumes 700 °C). The SRS process-safety-time run of SIF-02 shows the bed 1 exit still peaking at 744.1 °C with the SIF acting (791.8 °C without).	Either raise the pass-1 design temperature to 750 °C (thicker 304H shell, see EQ-em-05) or add the bed-1 inlet trip TSHH-201B recommended in the SRS so the exit stays near 631 °C; decide in the SRS/LOPA review.
EQ-em-02	SO3-V1-SA-HZP-001	B-101 duty quoted as ≈ 45 MW	HAZOP node 3 intent and the safety-layer register give B-101 ≈ 45 MW; the model gives 61.6 MW (94.0 t/h × (2801 − 440) kJ/kg), used on the datasheet.	Correct the HAZOP and register text to 61.6 MW; base the PSV-101 relief load (Relief & vent tab) on 94 t/h.
EQ-em-03	SO3-PID-001/002	P&ID rev A design-point table	P&ID table bed ΔP 4.5/3.0/3.6/2.9/3.0 kPa and stack 286 ppm; model PDI-204…244 4.52/2.86/3.16/2.54/2.50 kPa and AIT-306 292.5 ppm.	Update the table at rev B from the model (or the HMB SO3-V1-PE-HMB-001).
EQ-em-04	SO3-V1-ME-EQL-001	K-101 shaft power vs isentropic hand check	Model shaft power 4200 kW; isentropic power from 45 °C, −1 kPa g suction (assumed) to 60 kPa g is 3982 kW → implied isentropic efficiency 0.95 (typical 0.80–0.85, i.e. 4684–4977 kW). The model discharge 90 °C equals the isentropic outlet 91 °C.	Confirm suction temperature and efficiency with the blower vendor; size the motor on the vendor curve, not on the model power (see EQ-ee-01).

No.	Document	Subject	Finding	Proposed
EQ-em-05	SO3-V1-ME-EQL-001	R-201 shell thickness: pressure governs in pass 1	At 100 kPa g and 700 °C, 304H (S ≈ 27 MPa, E 0.85) needs 28.5 mm against a handling minimum of 15.5 mm; passes 2–5 need 11.0 mm (minimum governs). "Near-atmospheric, so the minimum governs" holds only below about 631 °C for this shell.	Zone the shell (pass-1 courses at the pass-1 design temperature, lower courses lower); consider internal insulation of pass 1 or a refractory-lined carbon-steel shell; confirm the design pressure basis (1.1 × blower shut-off).
EQ-em-06	SO3-PID-002	Second-pass line sizes	DN2000-PG-302 40 m/s; DN2000-PG-303 42 m/s; DN2100-PG-304 38 m/s at the design gas flow, against 20–25 m/s in P&ID general note 2.	Resize to about DN2700/DN2800 (22.5 m/s) or confirm the higher pressure drop against the K-101 head.